



إشارة

A Bilingual Sign Language Recognition & Communication System

Real-Time Letter and Word Recognition with a Web-Based Deployment

PRESENTED BY

Ahmed Adel Sayed Gouda · Hadeel Gamal Eldin Mohamed
Alaa Osama Mohammed Shehab · Salah Mohamed Salah Ibrahim

SUPERVISED BY

Dr. Nada Mostafa Abdelaziem
Dr. Reham Taher Al Magraby

MOTIVATION

The Communication Barrier

- **Sign language is the primary means of communication** for most deaf and hard-of-hearing people — not spoken language.
- **Human interpreters are effective but scarce**, costly, and cannot be present in every classroom, clinic, or service desk.
- **Automated recognition can scale** — yet most tools cover only one language, or only one recognition level.
- **Arabic Sign Language is under-served** despite large deaf populations across the Arab region.

466M

people worldwide live with disabling hearing loss

~7M

of them are in the Arab region

< 1 in 5

SLR research papers ship a usable, browser-accessible demo

What Existing Systems Miss

System / line	Lang.	Letters	Words	Live web
Typical WLASL work	EN	No	Yes	Rare
KArSL CNN-LSTM works	AR	No	Yes	No
ArASL2018 / ArSLA	AR	Yes	No	Limited
Eshara (this project)	EN + AR	Yes	Yes	Yes

Eshara fills a **structural gap**, not an accuracy gap:

one webcam-driven web product covering **both languages** at **both granularities** — four specialised models behind a single interface.

What We Set Out to Build

Aim Design, train, evaluate, and deploy a real-time bilingual ASL / ArSL system covering both letter and word recognition, accessible from a standard browser and webcam.

1

Four model pipelines

ASL & ArSL, letters & words

2

Per-model features

MediaPipe landmarks + RGB clips

3

Train / integrate models

MLP · Inception I3D · BiLSTM

4

Three-tier backend

Node gateway + FastAPI inference

5

React web client

live recognition + authenticated flow

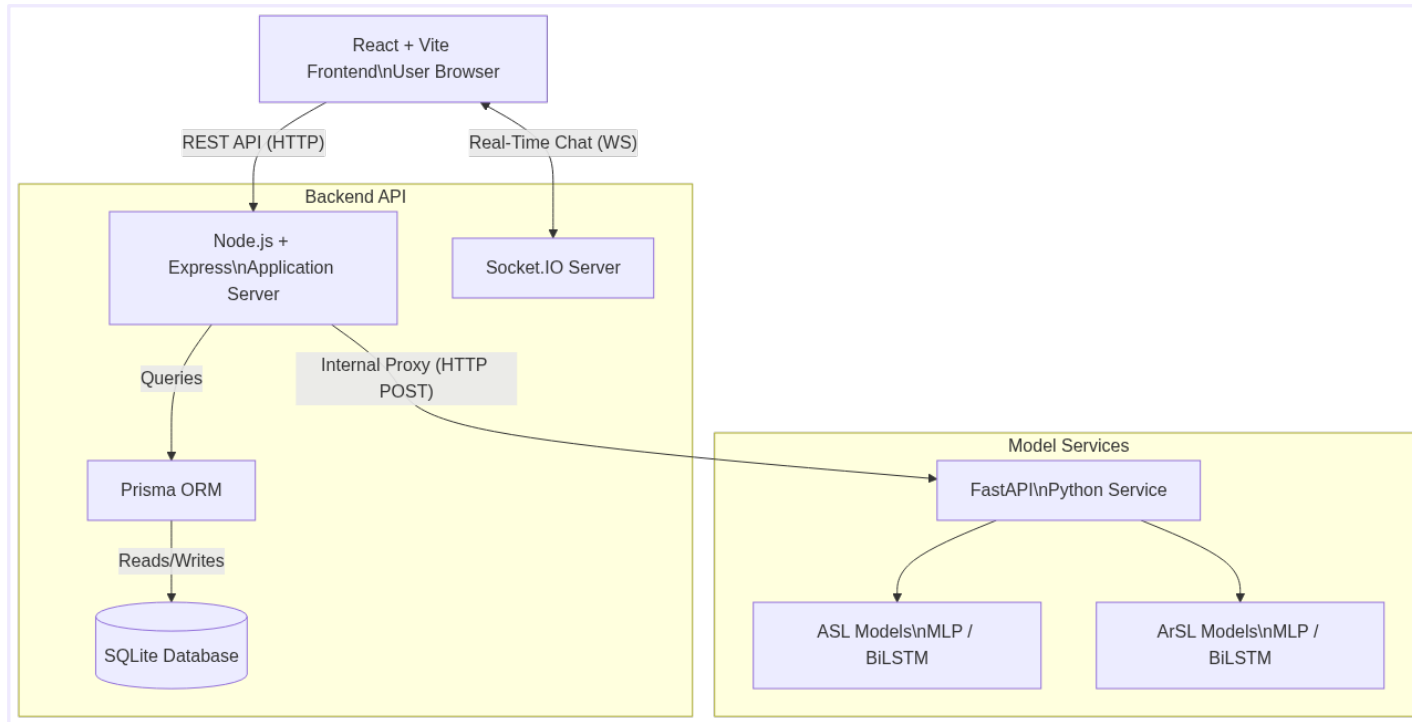
6

Rigorous evaluation

Top-1/Top-5, macro-F1, latency

Out of scope: continuous sentence translation, grammatical parsing, other sign languages, multi-signer scenes, native mobile apps, depth-camera hardware.

Three-Tier Web Architecture



Client Tier

React + Vite SPA

Captures webcam frames, renders skeletons & chat

Application Tier

Node.js + Express

JWT auth · Prisma/SQLite · Socket.IO · secure gateway

Inference Tier

FastAPI (Python)

Four models behind one /predict endpoint

Privacy by design: compact landmark vectors travel over the wire, not raw video — the FastAPI tier stays off the public internet.

One Model Matched to Each Task

Model	Task	Architecture	Input	Classes	Params	Top-1
ASL Letter	Letter	MLP	78-D hand	28	~49 K	99.06%
ArSL Letter	Letter	MLP	63-D hand	32	~183 K	99.63%
ASL Word	Word	Inception I3D	32× RGB clip	100	~12.5 M	65.89%
ArSL Word	Word	Stacked BiLSTM	48× 225-D pose	100	~255 K	99.62%

Static finger-spelling needs only per-frame pose → compact MLPs. **Word signs** need temporal modelling → 3-D CNN (RGB) for ASL, BiLSTM (landmarks) for ArSL.

No shared label space — bilingual capability = four parallel pipelines behind one API.

The Real-Time Letter Pipeline



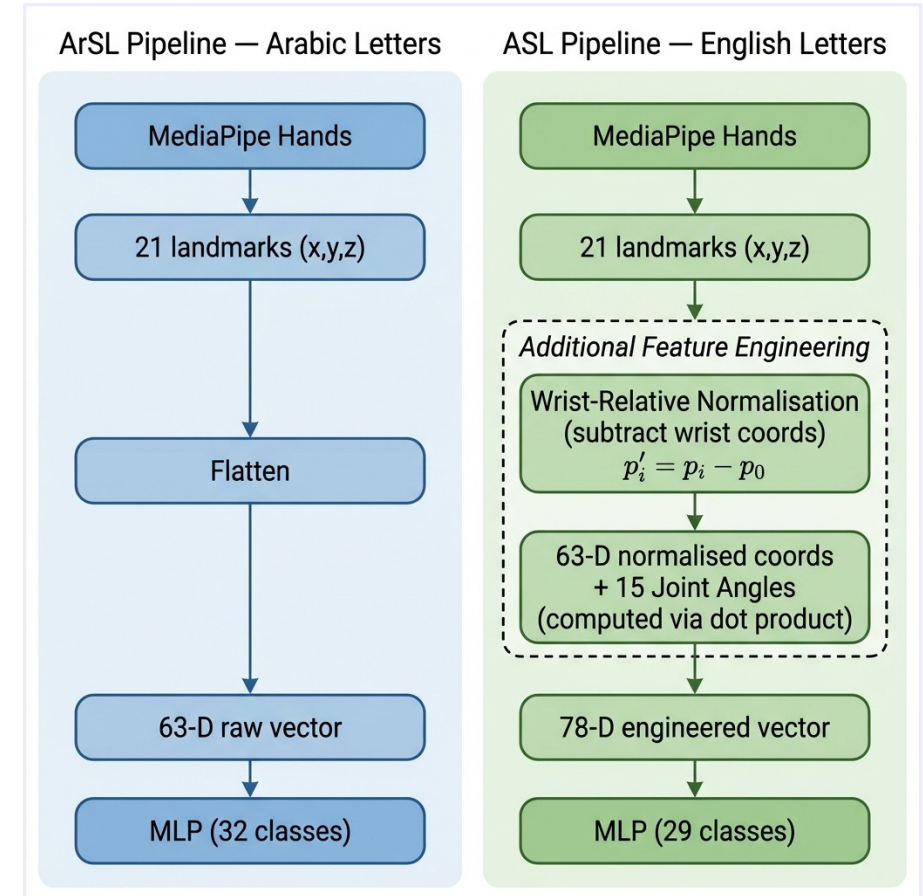
- **Seven sequential stages** turn each webcam frame into a committed letter, entirely in real time.
- **Frames are intentionally dropped** to keep latency low — not every frame is processed.
- **The model is pre-loaded** from a saved .h5 file before the camera loop ever starts.

“

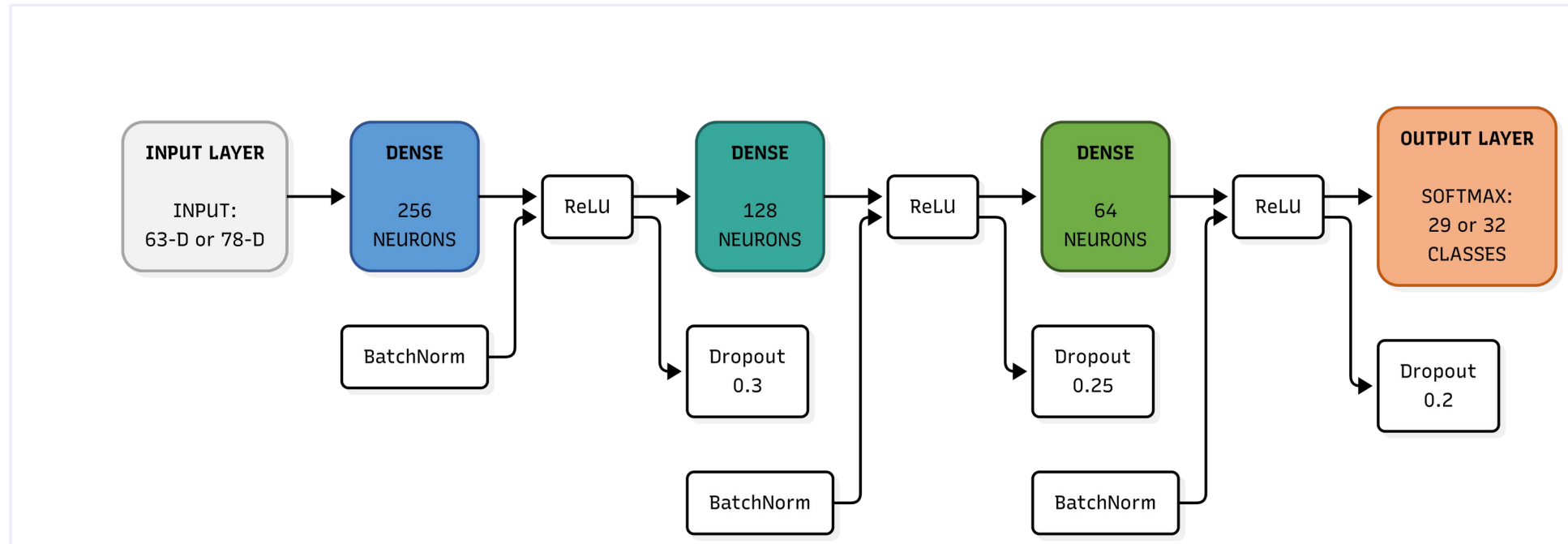
The system runs entirely in inference mode. No training happens at runtime.

From Landmarks to 78 Numbers

- **21 hand landmarks** from MediaPipe — 63 raw (x, y, z) values per frame.
- **Wrist-relative coordinates** — **63** features. Subtracting the wrist from every landmark removes any dependence on where the hand sits in the frame.
- **Joint bend angles** — **15** features. The angle at each finger joint, computed with the dot-product formula — scale-invariant, identical whether the hand is near or far.
- **63 + 15 = 78-D input vector** → fed straight into the MLP.



The MLP Classifier Architecture

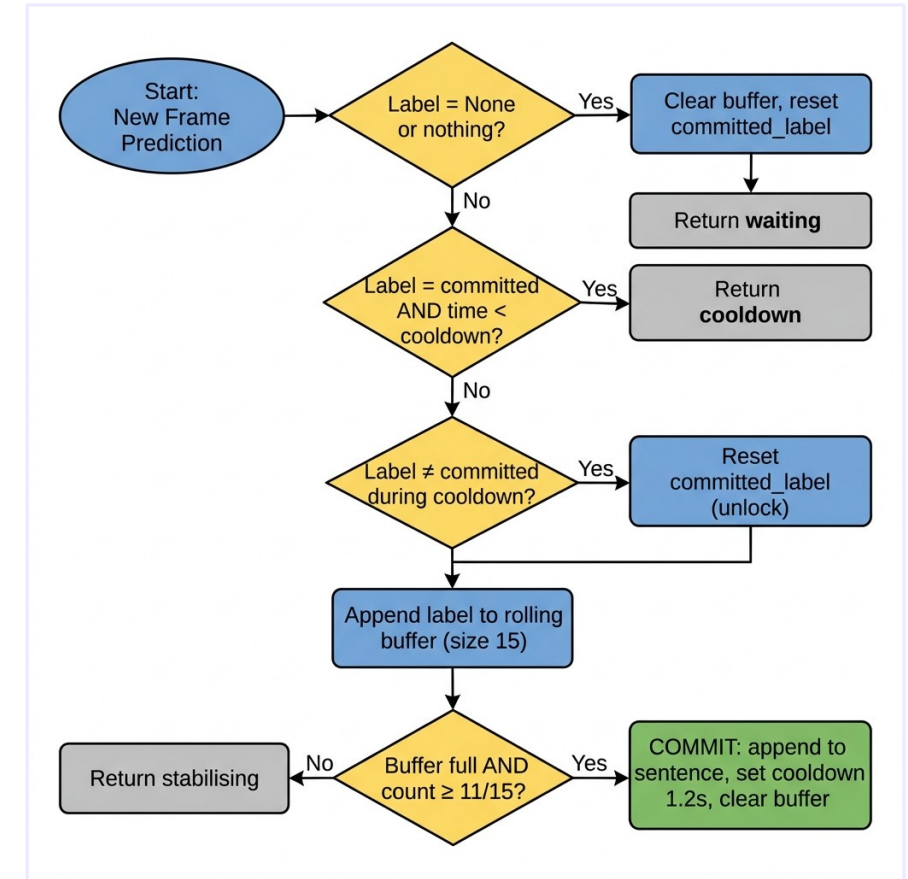


- **Input is exactly 78 numbers** — 63 position features + 15 angle features (the ArSL variant instead takes 63 raw landmarks).
- **Each hidden block applies BatchNorm → ReLU → Dropout** in that order, narrowing 256 → 128 → 64 neurons.
- **Softmax output** — 29 classes for ASL (A-Z plus space, del, nothing) and 32 for ArSL; 99.06% and 99.63% Top-1.

Committing Letters, Not Flicker

Turning noisy per-frame predictions into committed letters — a four-state tracker:

- **State 1 — No hand detected** : the tracker resets completely.
- **State 2 — Collecting votes** : the last 15 frame predictions are held in a rolling buffer.
- **State 3 — Commit** : triggered when 11 of 15 frames agree on the same letter.
- **State 4 — Cooldown** : output is locked for 1.2 s to prevent duplicate letters.
- **Confidence below 75%** is rejected before it ever enters the tracker.



Optimising for Real-Time Inference

Three implementation choices keep single-frame inference smooth at ~30 fps:

ThreadedCamera

A background thread reads the webcam continuously.

A size-1 queue keeps only the newest frame, so the main loop never waits — it always gets the most recent frame instantly.

tf.function Graph

Decorating inference with `tf.function` traces it once into a static graph.

Later calls skip Python interpreter overhead; a dummy warm-up call before the loop means frame one has no stutter.

Direct Model Call

Calling the model directly with `training=False` skips `model.predict`'s batching and callback overhead.

That matters when processing a single 1×78 input 30 times per second.

Word Models — Motion Over Time

ASL WORDS • MODEL 3

Inception I3D (3-D CNN)

- **Raw RGB clips** — 32 frames × 224×224, no landmark engineering
- **Inflated 3-D convolutions** learn spatio-temporal motion jointly
- **100 WLASL glosses**, ~12.5 M params (pretrained checkpoint)

65.89% Top-1 • 84.11% Top-5 (WLASL-100)

ArSL WORDS • MODEL 4

Stacked BiLSTM

- **225-D Holistic landmarks** — body + both hands, nose/wrist-normalised
- **48-frame sequences** read forward & backward
- **100 KArSL classes**, ~255 K params — lightweight for small data

99.62% Top-1 • 99.58% macro-F1 (KArSL)

Why two architectures? 3-D CNN absorbs signer/background variance from pixels; BiLSTM generalises from compact landmarks on a small corpus.

RESULTS

Held-Out Test Accuracy

99.06%

ASL Letters

MLP · 78-D · 28 classes

99.63%

ArSL Letters

MLP · 63-D · 32 classes

99.62%

ArSL Words

BiLSTM · 100 classes

65.89%

ASL Words

I3D · Top-1 (84.11% Top-5)

Reading the numbers honestly. The three landmark-based models exceed 99% because MediaPipe factors out signer appearance, background, and lighting before the classifier sees the data.

The ASL word I3D solves a genuinely harder problem — recognising motion directly from raw RGB pixels — so 65.89% Top-1 / 84.11% Top-5 reflects task difficulty, not a weaker model.

A Deployable Web Product

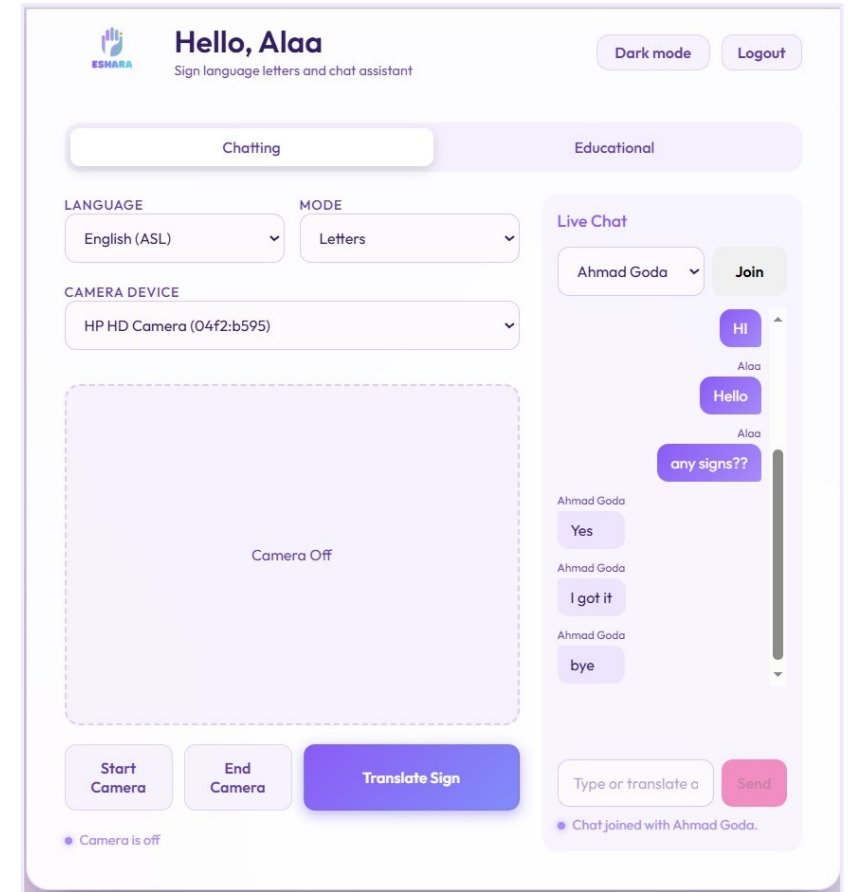
- **Live translation + chat** — select language & mode, sign to the webcam, send the prediction into a Socket.IO chat room.
- **Stateless JWT auth** with bcrypt hashing; Prisma ORM over SQLite for users, rooms & messages.
- **Encapsulated ML tier** — the FastAPI service is reachable only through the authenticated Node gateway.
- **Graceful degradation** — webcam denial, offline model service, and invalid sessions all fail safely.

51 ms

letter-mode end-to-end latency (p50) — well under the 200 ms target

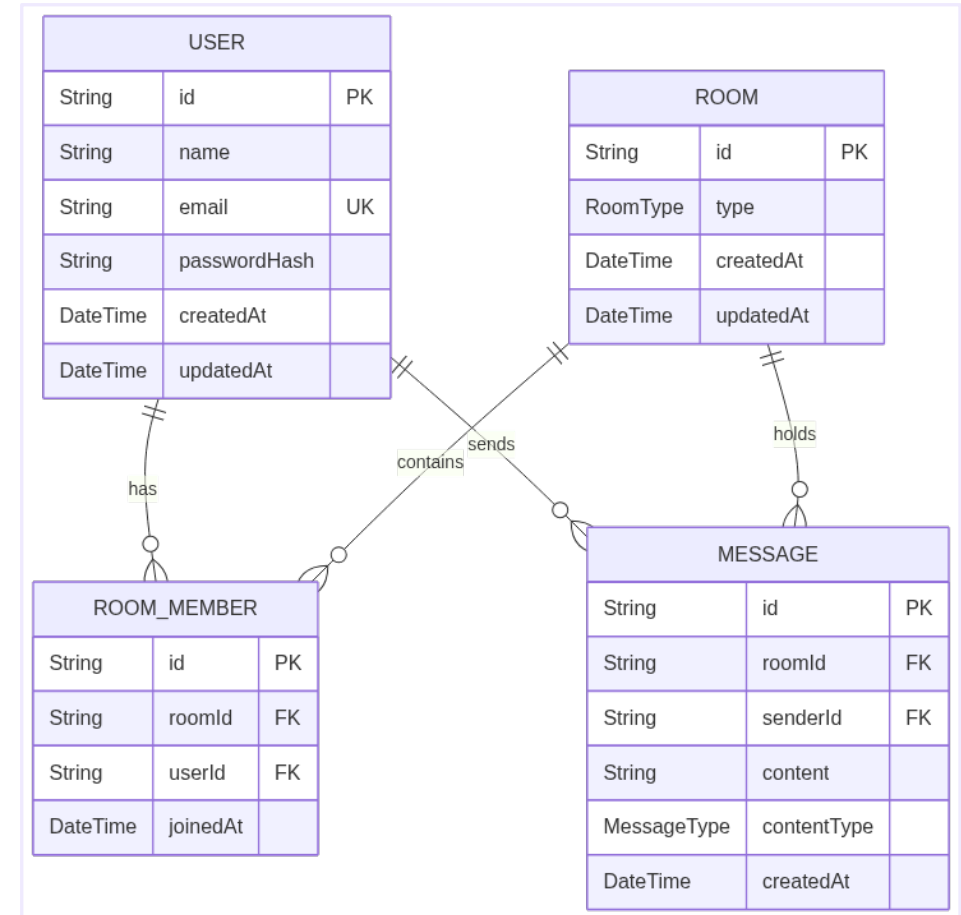
3 tiers

independently hostable: static SPA · Node server · FastAPI



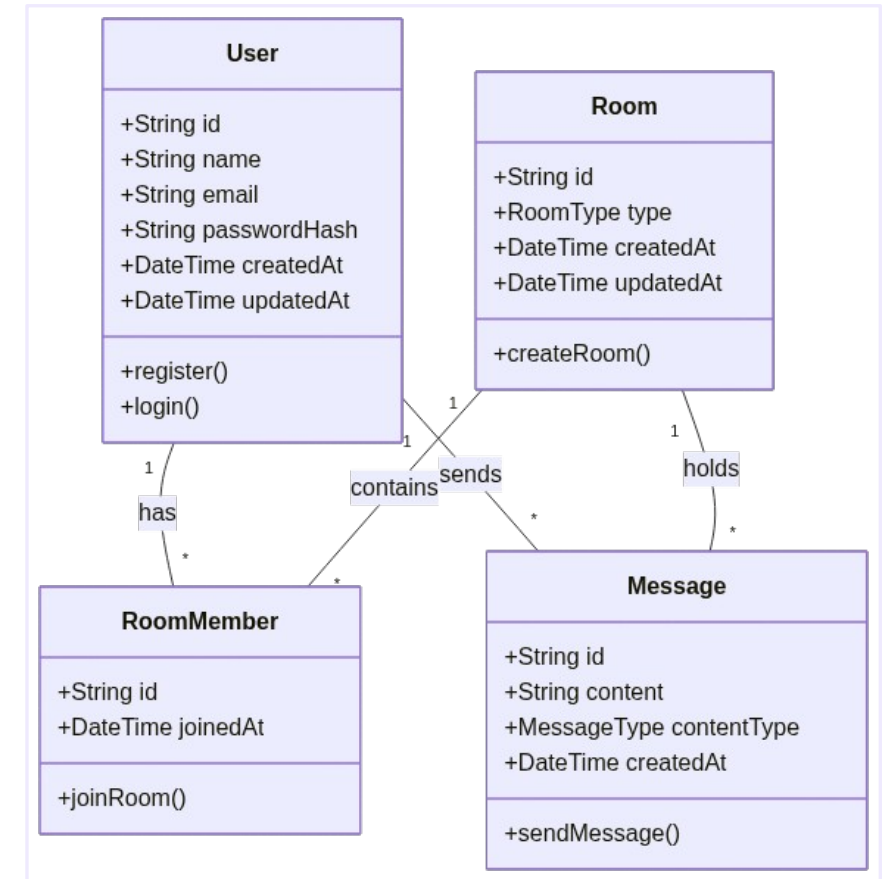
Data Model & Persistence

- **Prisma ORM** mediates every database access through a single typed schema (schema.prisma).
- **Relational storage** — SQLite in the current build, portable to Neon PostgreSQL for the cloud — centralized, secure and persistent.
- **Four core entities** — User, Room, RoomMember and Message.
- **Relationships** — users join one or more rooms; each room holds its members and the messages exchanged in it.



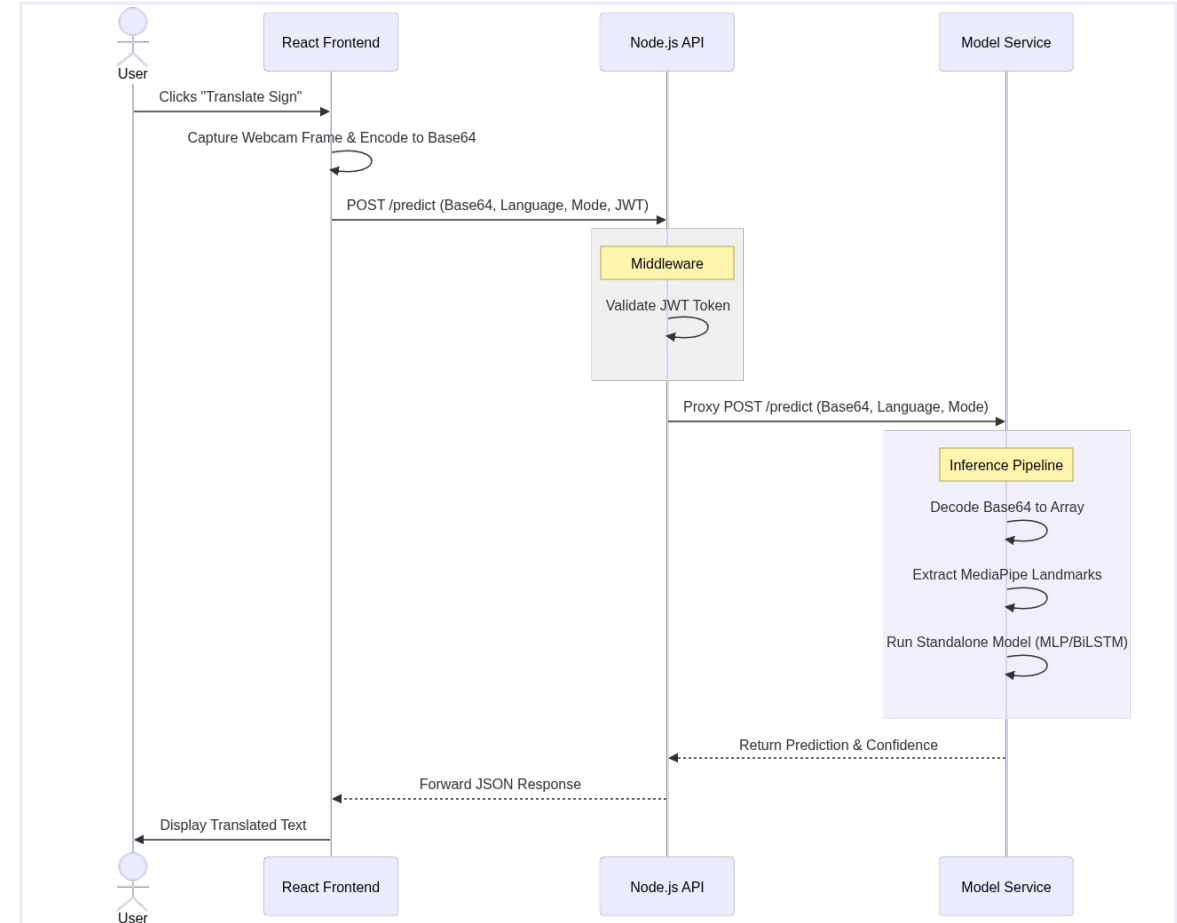
Class Diagram

- Each entity maps to a class with typed attributes and its own behaviour.
- **Key operations** — User.register() / login(), Room.createRoom(), RoomMember.joinRoom(), Message.sendMessage().
- **Associations mirror the ER model** — has, contains, sends and holds, with their multiplicities.
- **One schema, two views** — the class model and the ERD describe the same four objects.

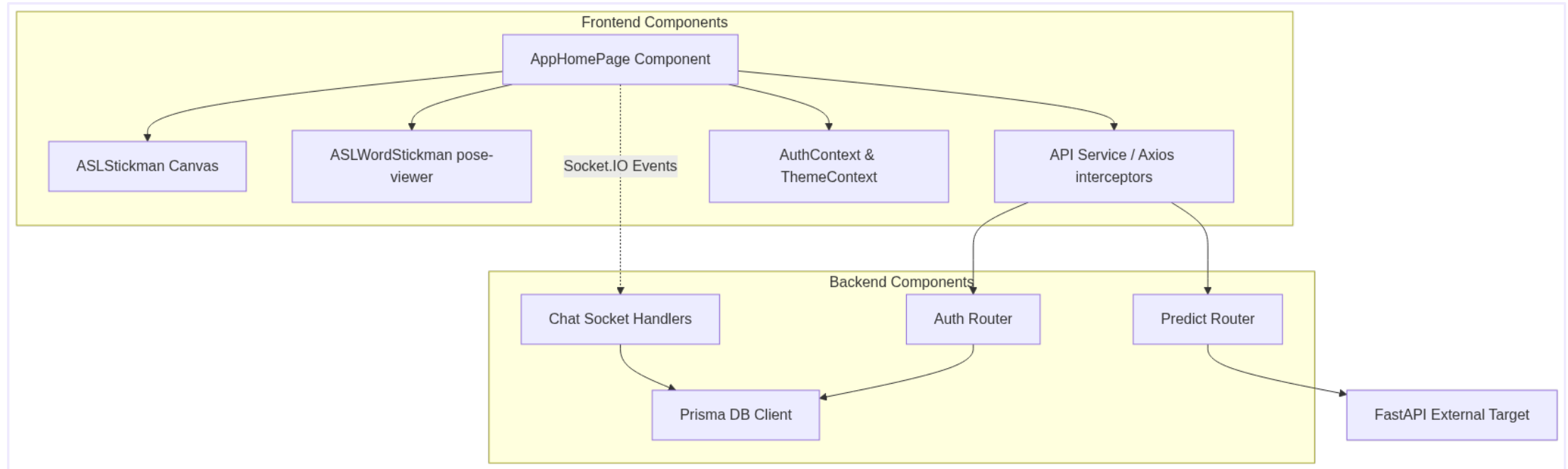


Prediction Sequence

- **Client** captures a webcam frame, encodes it to Base64 and POSTs /predict with language, mode and JWT.
- **Node middleware** validates the JWT, then proxies the request to the model service.
- **FastAPI pipeline** decodes the frame, extracts MediaPipe landmarks and runs the standalone model.
- **Response** — prediction + confidence flow back through Node to the UI as translated text.



Component Architecture



- **Frontend** — AppHomePage orchestrates the stickman canvas, pose-viewer, auth/theme context and the Axios API layer.
- **Backend** — Auth and Predict routers sit over the Prisma DB client, while chat socket handlers stream messages.
- **The Predict Router** is the sole gateway to the external FastAPI inference target.

The Education Module

Bilingual learning built on **motion-capture skeletons of real signers** — not stick figures. Three tools, in both ASL and ArSL:

Sign Viewer

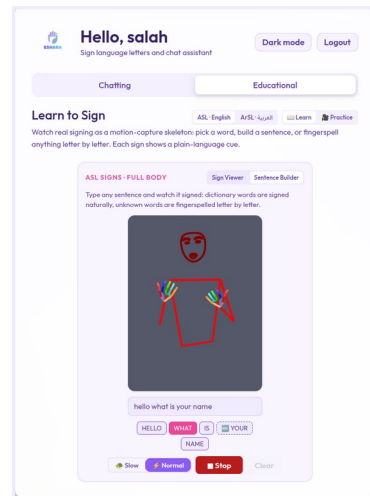
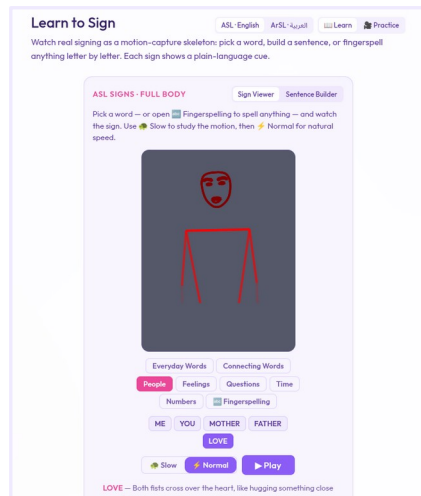
Browse a categorised vocabulary and watch each sign; fingerspell any typed word.

Sentence Builder

Type a free-form sentence — known words are signed, unknown words fingerspelled.

Practice Mode

Perform a prompted sign to the webcam and get instant feedback from the models.



- **Server-side animation stitching** joins many clips into one smooth signing animation.
- **44 EN + 42 AR verified word signs** across 9 categories, plus both full alphabets.
- **Content sourced from sign.mt** (ASL + Jordanian SL dictionaries), rendered by pose-viewer.

Guaranteeing Genuine Signs

The problem: the sign.mt dictionary silently FINGERSPELLS any word it lacks — with HTTP 200. A successful fetch does not prove a real sign exists, and publishing a spelled clip would teach learners something wrong.

1

Fetch the word AND the doubled word

the doubled form is guaranteed out-of-vocabulary → the cloud's own spelling

2

Reduce to handshape descriptors

wrist-relative, scale-normalised right-hand landmarks per frame

3

Compare with dynamic time warping

spelled fallback < 0.9 · genuine sign ≥ 2.9 — a clean separation

Result: every Arabic word in the app is a verified genuine sign; everything else fingerspells honestly. Bonus discoveries: the dictionary keys entries under proper orthography and definite-article forms.

Limitations & Failure Cases

MODEL & SCOPE

- **Isolated signs only** — the user pauses between signs; no continuous sentence decoding.
- **Dialect mismatch** — viewer teaches Jordanian SL, recognition trained on KArSL (Saudi).
- **Motion letters (J, Z)** — single-frame MLPs can't see the trajectory that defines them.

DATA & DEPLOYMENT

- **ASL words at 65.89%** — appearance-based recognition is inherently harder.
- **Signer generalisation** — accuracy drops on signers absent from training.
- **Constrained compute** — 4 GB GPU limited batch size and full-corpus training.

Positioning: *Eshara is an assistive & educational tool — a demonstration of feasibility, not a replacement for certified human interpretation.*

Future Work



Continuous recognition

Sentence-level SLR & translation on corpora like How2Sign.



Expanded Arabic vocabulary

More words & regional dialects, with deaf-community input.



Wire ArSL words to live web

Close the last integration gap in the deployed /predict path.



Mobile & edge

On-device inference via TensorFlow.js / ONNX Runtime.



Bidirectional communication

Pair sign-to-text with text-to-sign avatar synthesis.



Formal user studies

SUS evaluation with deaf & hard-of-hearing participants.

What Eshara Delivers



Unified bilingual web architecture

four heterogeneous SLR models served through one API



Full ASL + ArSL coverage

letters and words, with reproducible training pipelines



99.62% ArSL word recognition

a reliable 100-class Arabic word path on KArSL



Privacy-oriented inference

compact landmarks over the wire, not raw video



Verified bilingual education module

motion-capture signs guaranteed genuine, not fingerspelled



Deployable, low-latency product

browser + webcam only — 51 ms letter-mode latency

Eshara

إشارة

Thank You

Questions & Discussion

Ahmed Adel · Hadeel Gamal · Alaa Osama · Salah Mohamed

Supervised by Dr. Nada Mostafa & Dr. Reham Taher Al Magraby